

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Lab 10



Submitted to:

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Submitted by:

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Roll No: SP21379

Subject: SCD

Department: BSSE-5-Afternoon

Date: 14-12-2024

Tasks: 01

ID	Input	Expected output	Actual output	Class	Status
Case-01	The 4 hour workweek	Successful added	Successful added	Valid	Pass
Case-02	HW	Failed	Failed	Invalid	Pass
Case-03	456	Failed	Failed	Invalid	Pass
Case-04	The 4\$ hour workweek	Failed	Failed	Invalid	Pass
Case-05	Where Are the Children" by Mary Higgins Clarke (mystery) What Color is Your Parachute" by Richard N. Bolles (self-help) "Is Google Making Us Stupid?" by Nicholas Carr (thought-provoking work)	Failed	Failed	Invalid	Pass

Task: 02

Exception Handling Task

Arithmetic Exception

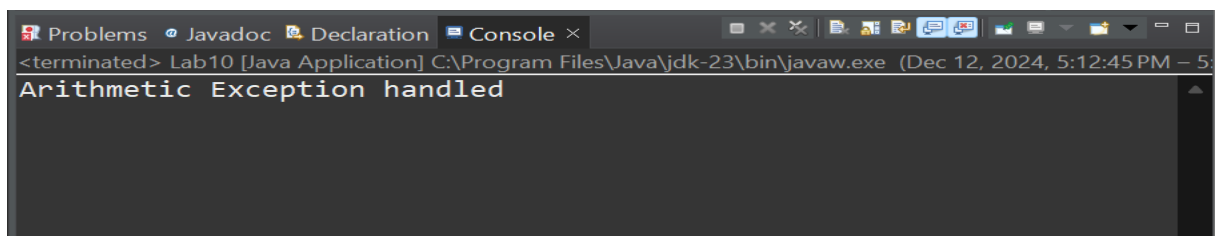
```
package lab10;

public class Lab10 {

    public static void main(String[] args) {
        // Arithmetic Exception
        try {
            int a=9;
            int b=0;
            System.out.println("division of a and b is= "+a/b);
        } catch (ArithmeticException e) {

            System.out.println("Arithmetic Exception handled ");
        }
    }
}
```

Output:



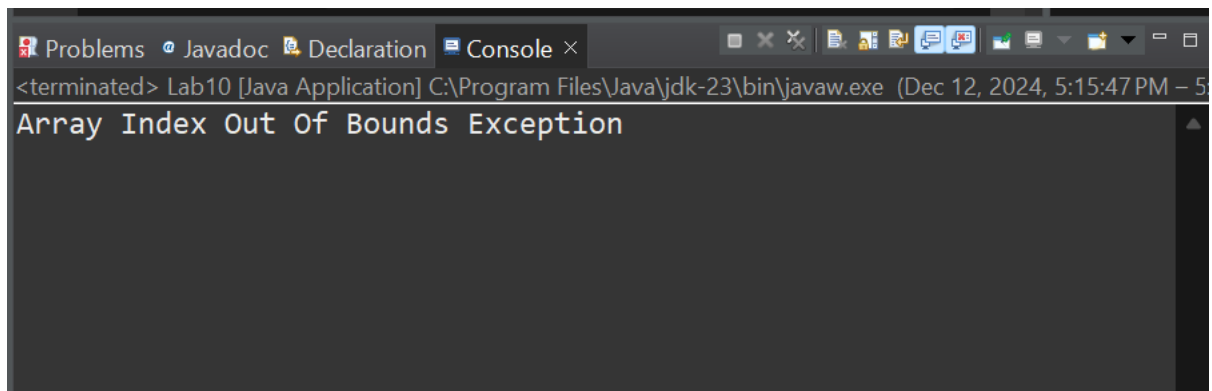
Array Index out of Bound Exception

```
package lab10;

public class task1 {
    static void validate(float salary) {
        //Array Index out of Bound Exception
        try {
            int[] arr=new int[10];
            System.out.println("value at 12 index = "+arr[12]);
        }catch(ArrayIndexOutOfBoundsException e) {

            System.out.println("Array Index Out Of Bounds Exception");
        }
    }
}
```

Output:

A screenshot of an IDE's console window. The title bar shows 'Problems', 'Javadoc', 'Declaration', and 'Console'. The console text reads: '<terminated> Lab10 [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (Dec 12, 2024, 5:15:47 PM - 5:'. Below this, the output of the program is displayed: 'Array Index Out Of Bounds Exception'.

Multiple catches

```
package lab10;

import java.util.InputMismatchException;

public class task3 {
    public static void main(String arg[]) {

        // Multiple catches
        try {
            int[] arr=new int[10];
            System.out.println("value at -8 index = "+arr[-8]);
        }
        catch(ArrayIndexOutOfBoundsException e) {

            System.out.println("exception handled"+e);
        }
    }
}
```

```

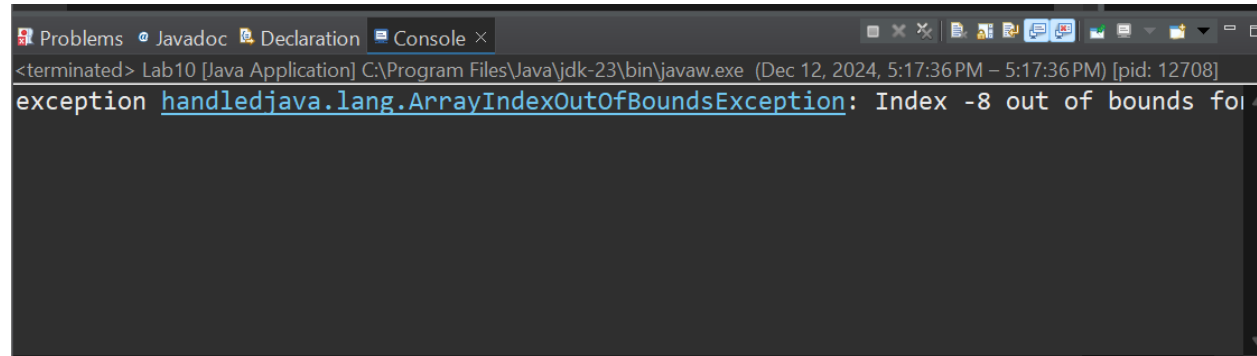
    }
    catch(InputMismatchException e) {

        System.out.println("exception handled"+e);
    }
    catch(Exception e) {

        System.out.println("exception handled"+e);
    }
}
}

```

Output:



The screenshot shows an IDE console window with the following text:

```

<terminated> Lab10 [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (Dec 12, 2024, 5:17:36 PM – 5:17:36 PM) [pid: 12708]
exception handledjava.lang.ArrayIndexOutOfBoundsException: Index -8 out of bounds for array of size 0

```

Input Mismatch Exception

```

package lab10;

import java.util.*;

public class task4 {

    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);

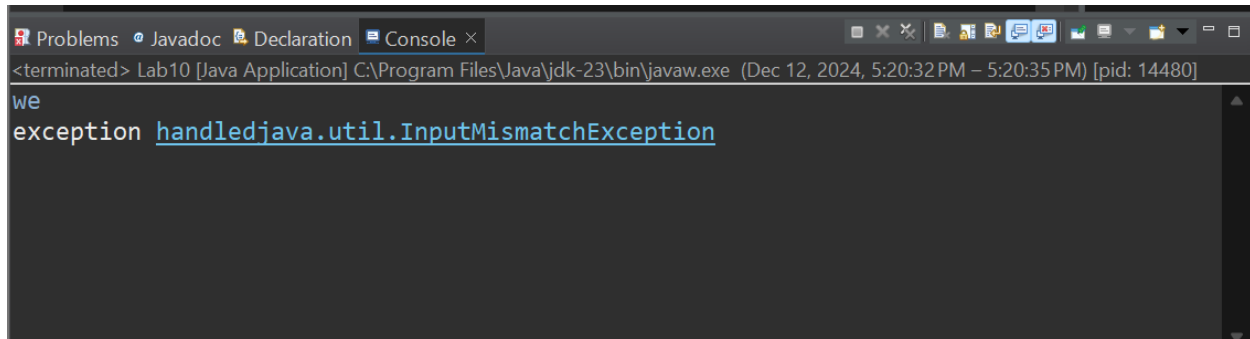
//Input Mismatch Exception

        try {
            int number=sc.nextInt();
            System.out.println("number is  = "+number);
        }
        catch(InputMismatchException e) {

            System.out.println("exception handled"+e);
        }
    }
    sc.close();
}
}

```

Output:

A screenshot of an IDE's console window. The title bar shows tabs for 'Problems', 'Javadoc', 'Declaration', and 'Console'. The console text reads: '<terminated> Lab10 [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (Dec 12, 2024, 5:20:32 PM – 5:20:35 PM) [pid: 14480] we exception handledjava.util.InputMismatchException'. The text is in a monospaced font with some color coding (blue for 'exception', green for the class name).

```
<terminated> Lab10 [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (Dec 12, 2024, 5:20:32 PM – 5:20:35 PM) [pid: 14480]
we
exception handledjava.util.InputMismatchException
```

Custom Exception

```
package lab10;
import java.util.Scanner;

class customException{
    static void validate(float salary) {
        if(salary<0) {
            throw new ArithmeticException("must be greater the
zero");
        }
        else {
            System.out.println("your salary is = "+salary);
        }
    }
}

public class task5 {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        System.out.println("enter your salary: ");
        float salary=sc.nextFloat();

        customException.validate(salary);

        sc.close();

    }
}
```

Output:

```
enter your salary:  
-40000  
Exception in thread "main" java.lang.ArithmeticException: must be greater the zero  
    at lab10/lab10.customException.validate(task5.java:7)  
    at lab10/lab10.task5.main(task5.java:28)
```